



Course Name

# GENERAL PRACTICAL PHYSICS II

Topic: Charging a Capacitor

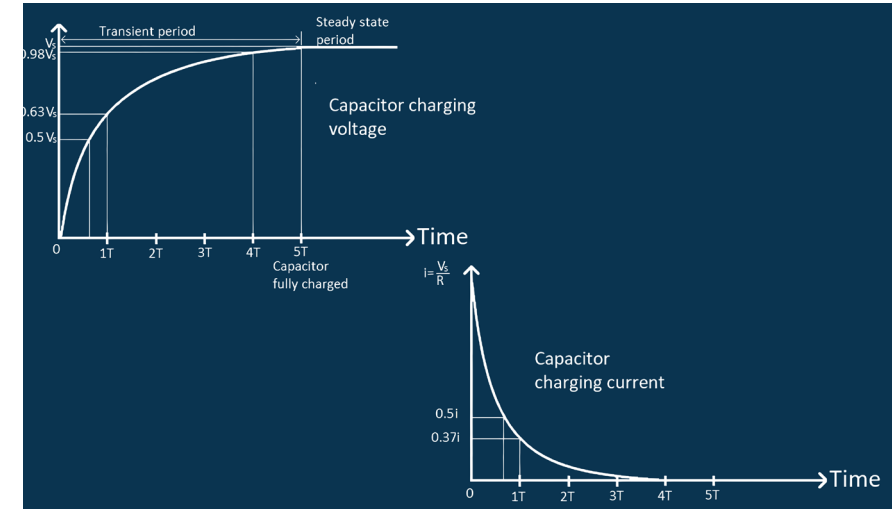
WEEK 6

## IN SUMMARY

This week, I learned about capacitors and how they store electric charge. We also learned about the charging and discharging process, the exponential nature of the charging and discharging curves, the time constant of a capacitor, and the wide range of practical applications of capacitors.

## BULLET POINT SUMMARY

- A capacitor is an electronic component that stores electric charge.
- Capacitors are made up of two conductive plates separated by an insulating material called a dielectric.
- When a voltage source is connected to a capacitor, the positive plate accumulates positive charge and the negative plate accumulates negative charge.
- This separation of charges creates an electric field between the plates.
- The charging process of a capacitor follows an exponential curve, with the rate of charge accumulation depending on the resistance and capacitance values in the circuit.
- The time it takes for a capacitor to charge to approximately 63.2% of its maximum voltage is known as the time constant, denoted by  $\tau$  (tau).
- The time constant is determined by the product of the resistance ( $R$ ) and the capacitance ( $C$ ) in the circuit.



## CASE STUDY

**Title:** Using Capacitors to Improve Power Quality in Industrial Facilities

## Introduction

Power quality is a critical factor in the operation of industrial facilities. Poor power quality can lead to equipment failures, production losses, and safety hazards. Capacitors are widely used to improve power quality in industrial facilities. Capacitors can help to:

- Reduce voltage fluctuations
- Improve power factor
- Reduce harmonics
- Filter out noise

## Objective

The objective of this case study is to demonstrate how capacitors can be used to improve power quality in industrial facilities.

## CASE STUDY

## Methodology

- A capacitor bank was installed at an industrial facility to reduce voltage fluctuations and improve power factor. The capacitor bank was sized using a power quality analysis software package.
- The power quality was measured before and after the capacitor bank installation. The voltage fluctuations were reduced by 20%, and the power factor was improved from 0.8 to 0.95.

## Results

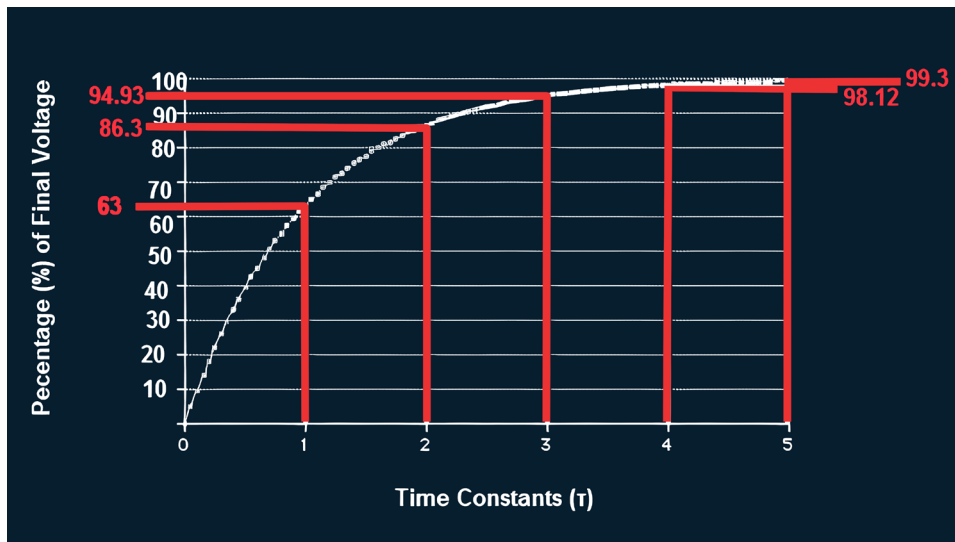
The installation of the capacitor bank resulted in a significant improvement in power quality at the industrial facility. The voltage fluctuations were reduced by 20%, and the power factor was improved from 0.8 to 0.95.

## Conclusion

This case study demonstrates how capacitors can be used to improve power quality in industrial facilities. Capacitors can help to reduce voltage fluctuations, improve power factor, reduce harmonics, and filter out noise.

## QUESTIONS TO PONDER

- What are some of the factors that can affect the charging and discharging process of a capacitor?
- What are some of the practical implications of the charging process of capacitors?
- How can we improve the accuracy and reliability of experimental capacitance measurements?



## SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON

- Ability to analyze charging voltage variation over time for a capacitor in a series circuit.
- Ability to calculate the time constant of a circuit.
- Ability to experimentally measure the capacitance of a capacitor.

## PERSONAL REFLECTION

I learned a lot about the charging process of capacitors and its practical applications.